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# Effect of the Targeted Cluster Initiative on malaria indicators in Ethiopia

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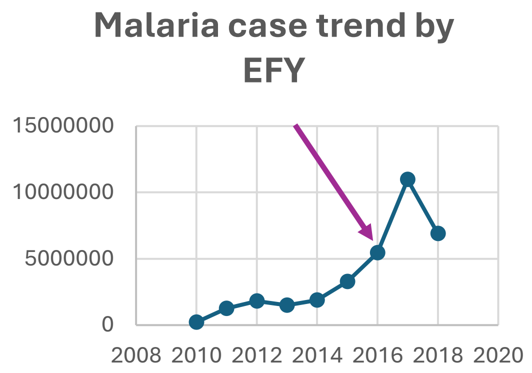
# Background: Malaria in Ethiopia




- ~75% of Ethiopia is malarious; 69% of the population is at risk of malaria.
- Major progress was achieved through sustained malaria control efforts (LLINs, IRS, early diagnosis, and prompt treatment) - reduced reported cases to ~0.9 million in 2019.
- Despite two decades of significant progress, malaria has resurged since 2022: cases increased from 1.3M cases (2021) → 3.3M (2022) → 4.1M (2023)
- The resurgence intensified and expanded in 2024, with an estimated 12.4M malaria cases, affecting more areas and persisting beyond expected seasonal peaks.
- It has been driven by multiple factors, including:
    - Climate and environmental changes
    - Conflict-related health-system disruptions and programmatic challenges
    - Population displacement and other population movements
    - Biological factors and changes in malaria vectors/parasites

# What is The Targeted Cluster Initiative?

- A six-month intensive malaria response launched by the Ministry of Health in June 2024 in response to the malaria resurgence.
- Targeted **222 high-burden districts**, representing **~76%** of Ethiopia's malaria burden in 2024 while accounting for only 21% of all districts.
- Districts were organized into **9 operational clusters**, considering malaria burden, geographic accessibility, and conflict context.
- The initiative aimed to rapidly reduce malaria morbidity and mortality, particularly in high-burden and conflict-affected areas.
- 100+ technical experts from government institutions, agencies, and implementing partners were deployed across the clusters.

# How was it implemented?

- TCI intensified existing malaria interventions through coordinated action

**1. Intensified case detection & management**

- Expanded malaria testing and active/passive case detection
- Reactivated inactive Health Posts to improve access to diagnosis and treatment
- Strengthened HEW engagement for early diagnosis, referral, and prompt treatment

**2. Strengthened vector control**

- Intensified larval source management
- Conducted environmental management to reduce mosquito breeding sites

**3. Community engagement**

- Conducted house-to-house HEW visits to increase awareness of malaria symptoms and promote early care-seeking
- Promoted LLIN utilization through SBC and household-level education
- Used community outreach and mass media to reinforce malaria prevention messages

**4. Data-driven coordination & accountability**

- Conducted weekly surveillance data reviews and regular performance reviews
- Used data to identify implementation gaps and guide corrective action
- Strengthened multi-stakeholder coordination, supportive supervision, and accountability

- **National leadership & coordination:** Ministry of Health
- **Technical & operational coordination:** FMoH + Agencies + partners

# Why evaluate the impact of the Targeted Cluster Initiative?

A major national response should leave an evidence base.

TCI was a major national malaria response

However, a consolidated and accessible evaluation of the TCI experience was not readily available.

Documenting the TCI experience can provide a reference for future malaria responses and similar targeted interventions.

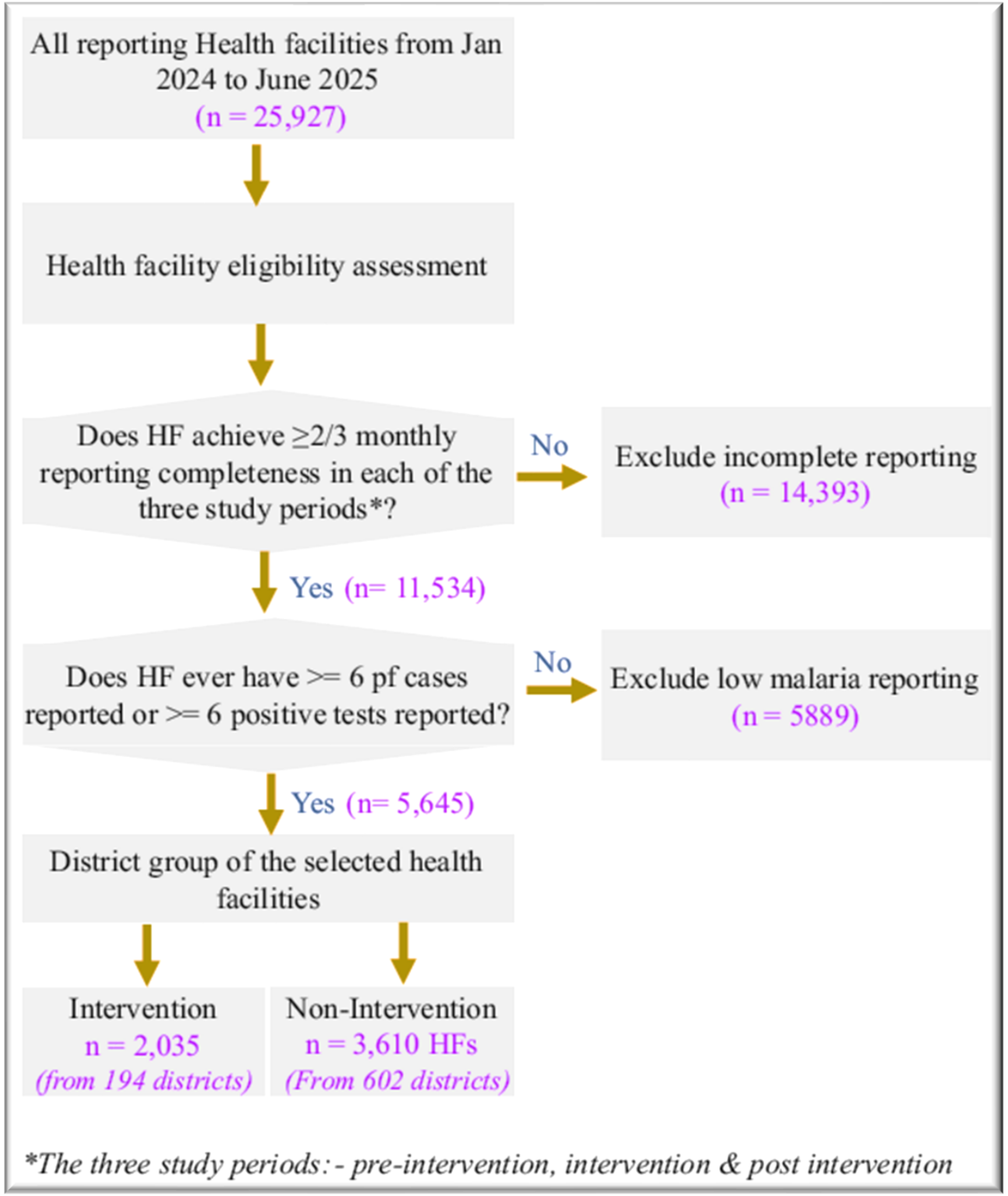
Our aim: Assess what changed during and after TCI and generate an accessible evidence base for future use

# Method

- **Study design:**
  - **A non-randomized controlled before-and-after (CBA) design**, a quasi-experimental approach based on routine HMIS data.
- **Study period:**
  - 18 months | Jan 2024 – Jun 2025

| **Period** | **Definition** |
|---|---|
| Jan–Jun 2024 | Pre-intervention |
| Jul–Dec 2024 | TCI implementation |
| Jan–Jun 2025 | Post-intervention |

- TCI districts ↔ Non-TCI districts
  - Malaria indicators were compared across the three periods and between the two study arms



# Cont….

- The analysis covered 5,645 health facilities across 796 districts
  - Intervention districts (n= 194, pop = 19.1M)
    - ✓ 2,035 facilities
    - ✓ 35% of reporting facilities
    - ✓ ~80% of reported cases
  - Non-intervention districts (n= 602, pop = 72.3 M)
    - ✓ 3,610 facilities
    - ✓ 18% of reporting facilities
    - ✓ ~75% of reported cases

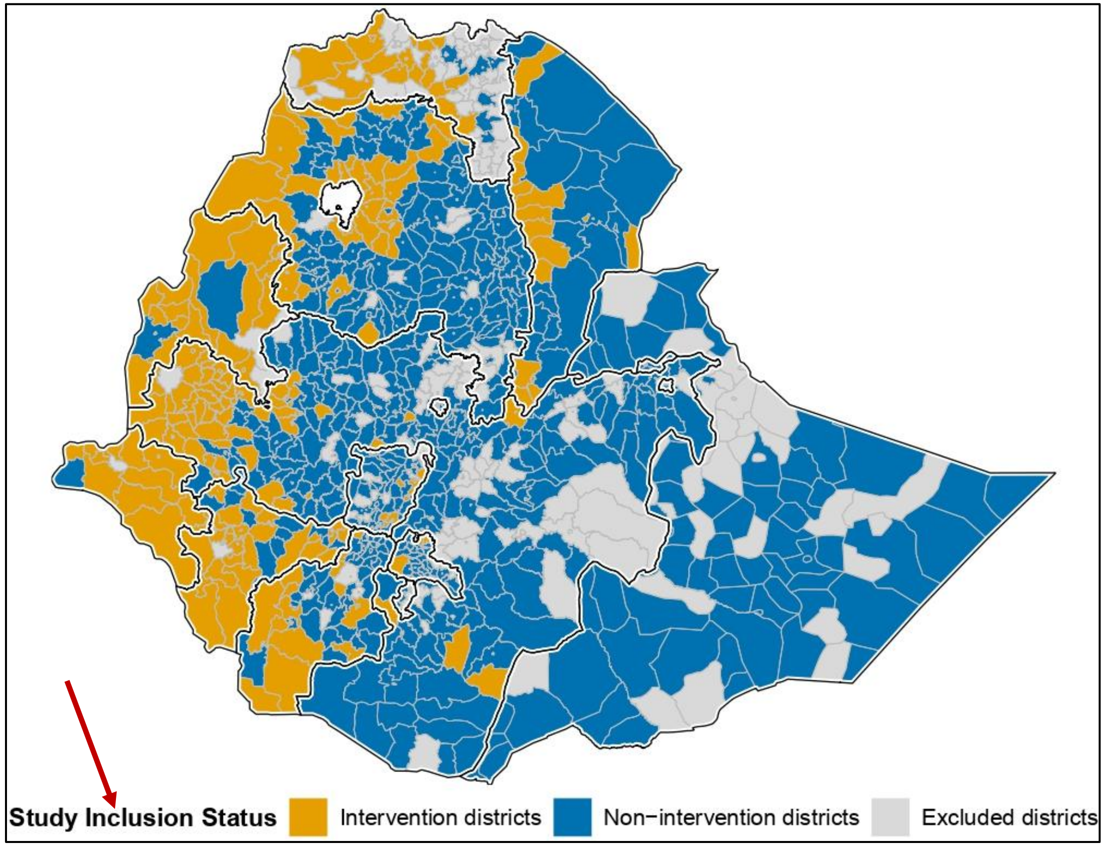
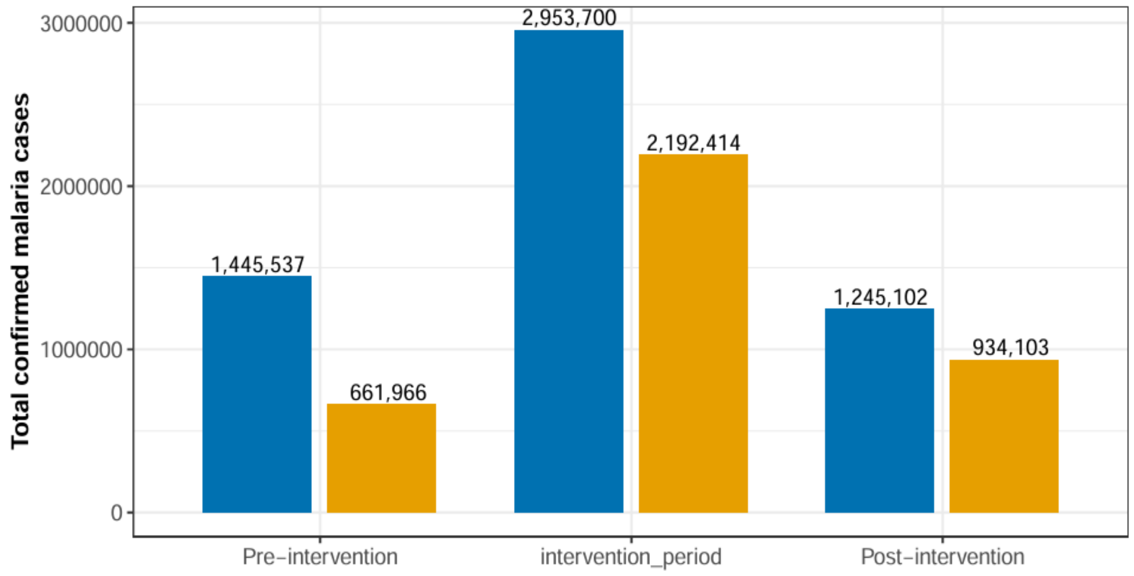


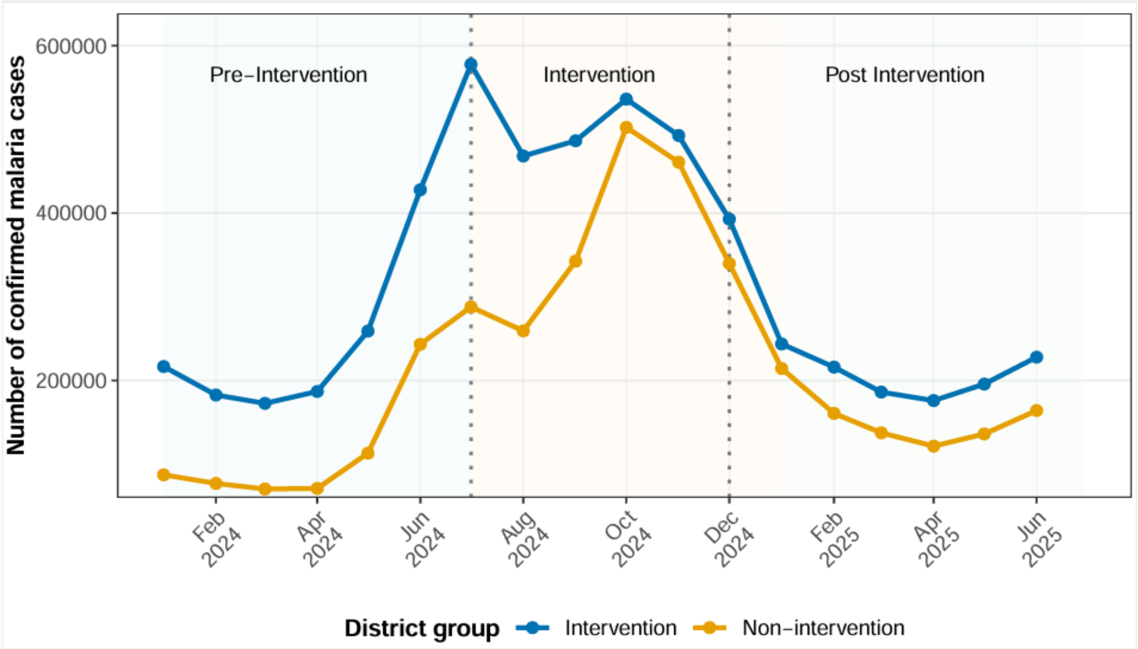

***Figure**: Geographic distribution of study districts by intervention status*

# Results: *confirmed cases*

- 9.43 million confirmed cases were reported during the 18-month study period: 5.64M (**60%**) in TCI districts and 3.79M (40%) in non-TCI districts.
- Confirmed cases increased during TCI implementation in both TCI and non-TCI districts.
- Cases declined after TCI ended in both groups, with the share contributed by TCI districts decreasing from **69% pre-TCI** to **57% post-TCI.**
- Monthly trends broadly followed Ethiopia's seasonal transmission pattern, with peaks during the major (Sep–Dec) and minor (Apr–Jun) transmission seasons.

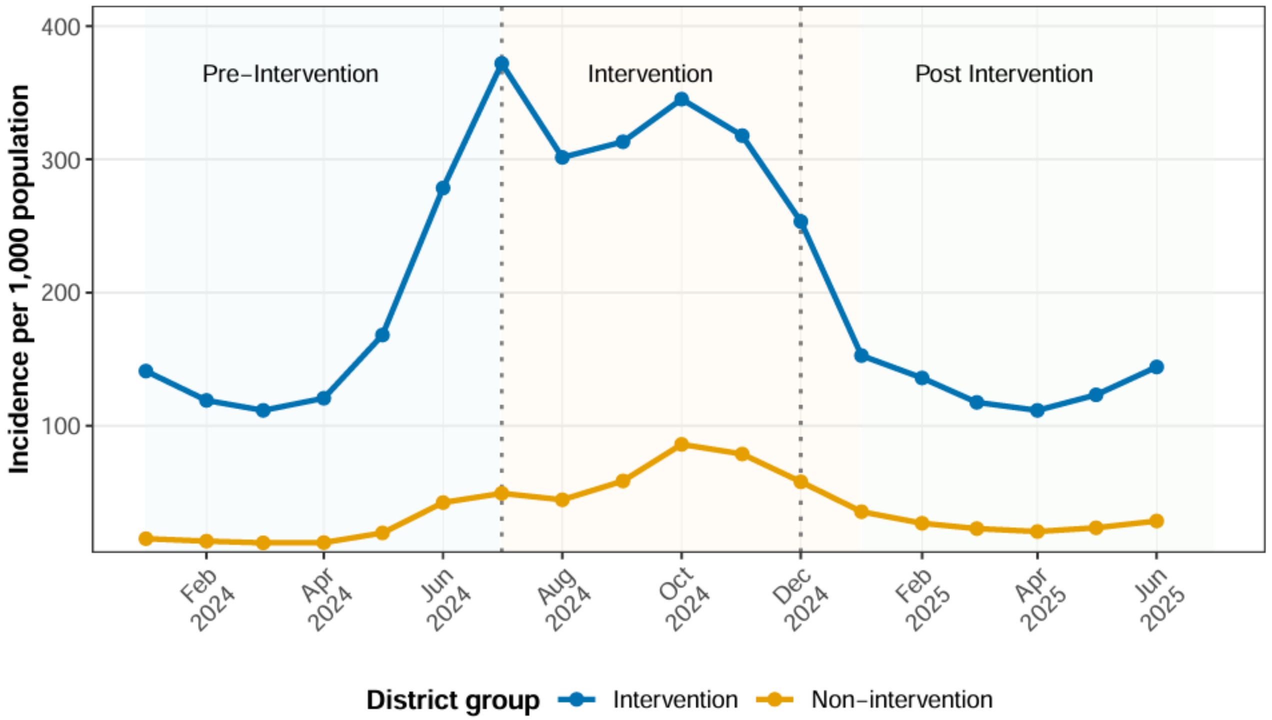
Following the intervention period, malaria cases declined in both groups, with a greater relative shift in the reported burden away from TCI districts. This pattern is consistent with improved malaria control in targeted districts, although seasonal transmission and other factors may also have contributed.



# **Results:** ***Malaria Incidence Rate***

- Incidence increased during TCI implementation in both groups:
  - **TCI:** 155 → **317**
  - **Non-TCI:** 19.0 → **62.2**
- Post-TCI incidence declined to:
  - **TCI: 130** per 1,000
  - **Non-TCI: 25.8** per 1,000
- **Change from baseline:**
  - **TCI: −16.1%**
  - **Non-TCI: +35.8%**
- Monthly trends broadly followed **seasonal transmission patterns**.



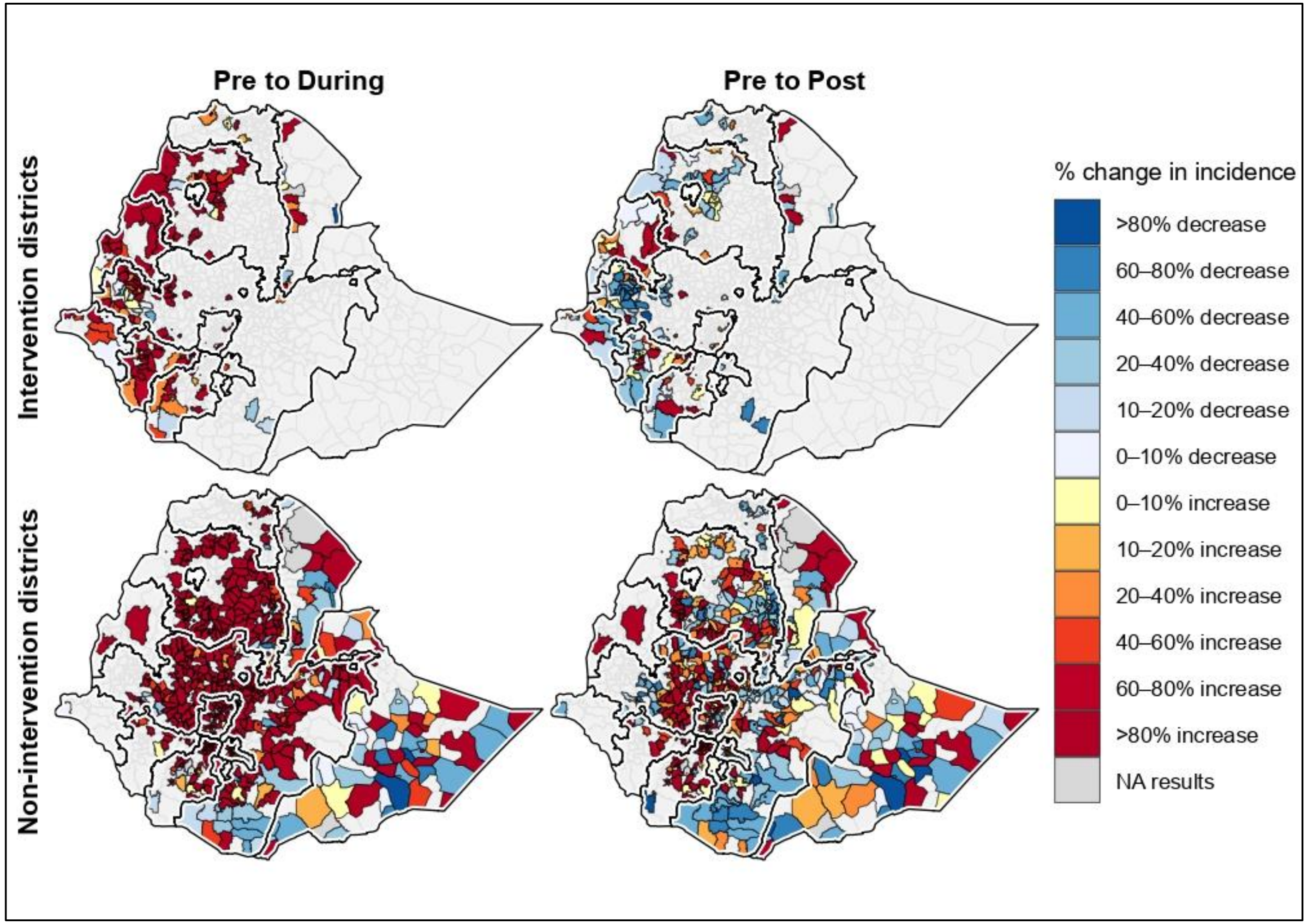

Despite a seasonal increase during TCI implementation, post-intervention incidence fell below baseline in TCI districts but remained above baseline in non-TCI districts.

# Incidence cont….

District-level change in incidence: Pre- vs Post-TCI

- **TCI districts**
  - **59%** of districts showed a reduction in incidence
- **Non-TCI districts**
  - **45%** of districts showed a reduction in incidence

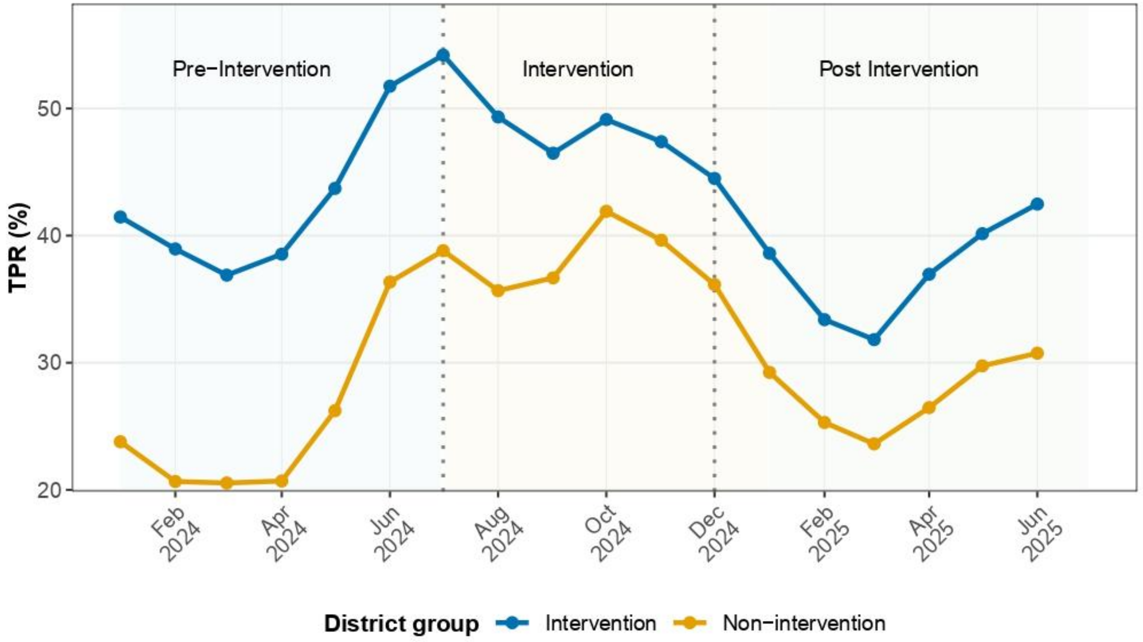
***Figure:** Percentage change in annualized malaria incidence across TCI periods in intervention and non-intervention districts.*




A greater proportion of TCI districts experienced a reduction in incidence from pre- to post-intervention compared with non-TCI districts, although district-level changes varied substantially.

# Results: *Test positivity rate*

- **During TCI:** TPR increased in both groups
  - TCI: **42% → 49%**
  - Non-TCI: **25% → 38%**
- **Post-TCI:** TPR declined to
  - TCI: 37% (**12% below baseline**)
  - Non-TCI: 28% (**12% above baseline**)
- Monthly TPR patterns broadly followed **seasonal transmission patterns**.




**NB:** TPR is a facility-based indicator and may be influenced by testing practices, diagnostic availability, healthcare-seeking, and facility utilization.

TPR declined below baseline in TCI districts, alongside the observed reduction in incidence, indicating a more favorable post-intervention trajectory than in non-TCI districts.

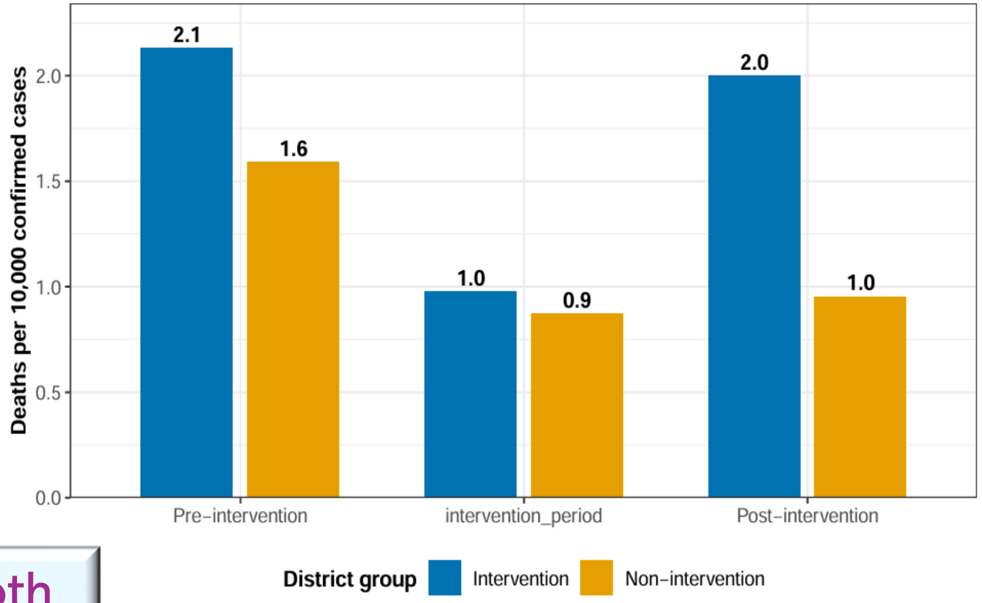
# Results: *Indicators of severe malaria outcomes*

**Malaria-related admissions**

- Admission rate declined in **both groups** across the study periods.
  - ✓ **TCI:** 2.7 → 1.58 → 1.72 per 100 confirmed cases
  - ✓ **Non-TCI:** 2.4 → 1.4 → 1.3 per 100 confirmed cases
- Post-intervention vs baseline: **−36.3% TCI; −45.8% non-TCI**

**Case fatality rate (CFR)**

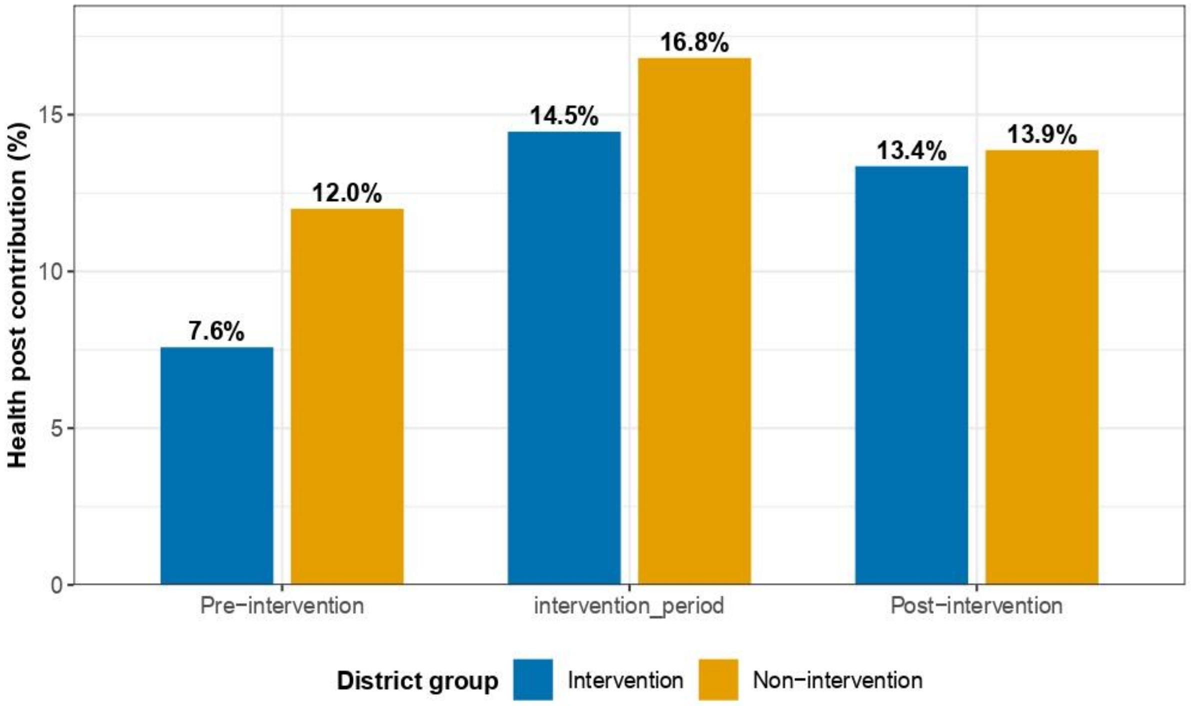
- CFR decreased during the intervention period in both groups:
  - ✓ **TCI**: 0.0213 → 0.00975
  - ✓ **Non-TCI:** 0.016 → 0.009
- Post-intervention: CFR in TCI districts returned close to baseline, while it remained below baseline in non-TCI districts.




Admissions and CFR declined during the intervention period in both groups. These outcomes may be influenced by changes in case management, healthcare-seeking, referral, and reporting practices.

# Results: *Health Post contribution to malaria case detection*

- Contribution of Health Posts increased in **both groups** after TCI implementation.
  - ✓ **TCI districts:** 7.6% → **13.4%** **+76.3% relative increase**
  - ✓ **Non-TCI districts:** 12% → **14%** **+16% relative increase**
- Health Posts continued to contribute a slightly higher share of cases in non-TCI districts, but the increase from baseline was substantially greater in TCI districts.




The larger increase in Health Post contribution in TCI districts suggests strengthened community-level malaria detection during the intensified response.

# Summary of Results

| Indicator | District Group | Pre-intervention | During intervention | Post-intervention | % change Comparing pre and post |
|---|---|---|---|---|---|
| Incidence (/1000) | Intervention | 155 | 317 | 130 | ↓ **16.1%** |
| Incidence (/1000) | Non-intervention | 19 | 62.2 | 25.8 | ↑**35.8%** |
| TPR (%) | Intervention | 42% | 49% | 37% | ↓ **11.9%** |
| TPR (%) | Non-intervention | 25% | 38% | 28% | ↑**12%** |
| CFR | Intervention | 0.0213 | 0.0098 | 0.02 | ↓ **6%** |
| CFR | Non-intervention | 0.016 | 0.009 | 0.01 | ↓ **37.5%** |
| Admissions (%) | Intervention | 2.7% | 1.58% | 1.72% | ↓ **36.3%** |
| Admissions (%) | Non-intervention | 2.4% | 1.4% | 1.3% | ↓ **45.8%** |
| HP contribution (%) | Intervention | 7.6% | 14.5% | 13.4% | ↑**76%** |
| HP contribution (%) | Non-intervention | 12% | 16.8% | 13.9% | ↑**15.8%** |

Overall, TCI districts showed more favorable post-intervention trends in incidence and TPR, alongside a larger increase in Health Post contribution to case detection.

# Key limitations of the study

### 1. Routine HMIS data quality

✓ Incomplete reporting, reporting errors, and variability in data quality across facilities and districts may have affected the findings.

### 2. Non-equivalent study groups

✓ TCI districts were **purposively selected for high malaria burden and operational vulnerability**, limiting direct comparability with non-TCI districts.

### 3. Seasonal and contextual influences

✓ The TCI period overlapped with the major malaria transmission season, while climate, population movement, conflict, and other contextual factors may also have influenced observed trends.

### 4. Causal inference is limited

✓ The pre–post design cannot establish that observed changes were caused by TCI.

Despite these limitations, the analysis provides valuable operational evidence from a large-scale national malaria response and identifies patterns that can inform future response strategies.

# Conclusion

- TCI districts experienced a more favorable post-intervention malaria trajectory, with incidence and TPR declining below pre-intervention levels.
- Community-level case detection increased more substantially in TCI districts, reflected by the greater increase in Health Post contribution.
- Findings are consistent with a potential contribution of TCI to improved malaria control, although causal attribution is not possible.
- The evaluation provides an evidence base to inform and refine future targeted malaria response strategies.

- Overall, the post-intervention pattern was more favorable in TCI districts than in non-TCI districts.

# Recommendations

## Evaluate future large-scale malaria responses

- Generate accessible evidence and document implementation lessons for future response planning.

## Consider targeted, cluster-based approaches

- Use targeted approaches where malaria burden and operational vulnerability are concentrated.

## Strengthen community-level detection and response

- Build on Health Posts and community-based surveillance to improve early detection and response.

## Improve HMIS data quality and standardization

- Strengthen reporting completeness and consistency to support reliable monitoring and evaluation.

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